

FastPAYS

Problem Statement: Instant Service Booking System

Domain: Web Development (On-Demand Services)

Description: At present, it seems that people don't have a platform where they can book household emergency or quick, dispatch services. Most of the existing service directories are unverified, and the traditional platforms for booking usually work on scheduled time slots rather than on immediate service delivery. A system that not only guarantees a rapid response time but also offers real, time visibility on arrival estimates is needed.

Objectives:

1. Primary Objective: Create a platform where users can book house help services and receive assistance within 15 minutes.
2. Automation: Implement smart automation that instantly matches a booking request with the nearest idle service provider.
3. Transparency: Provide real-time tracking and accurate Estimated Time of Arrival (ETA).
4. Ease of Use: To simplify the user interface so that a booking can be completed in under 5 clicks.

Scope: The project will cover the following modules and functionalities:

1. Functional Requirements:
 1. User Registration & Authentication
 2. Service Category Selection: clearly defined categories like Plumber, Electrician, Cleaner, Cook, etc.
 3. Instant Booking Engine:
 - a. User selects a service.
 - b. System scans for providers within a 2 km radius.
 - c. Auto-assigns the job to the nearest available provider.
 4. Booking Confirmation: Generation of a unique Booking ID immediately upon assignment.
 5. ETA Display: Calculation of arrival time based on the provider's current GPS location and traffic conditions.
 6. Booking History: A dashboard for users to view past services, payments, and invoices.
2. Non-Functional Requirements:
 1. Performance: The matchmaking algorithm must execute quickly.
 2. Scalability: The database should handle concurrent booking requests.
 3. Reliability: 99.9% uptime to ensure emergency services are always accessible.

Proposed System Logic: To achieve the 15-minute objective, the system uses a Proximity-First Algorithm:

1. Request Initiation: User clicks "Book Now."
2. Radius Scan: The backend scans the database for service providers who are marked "Active" and are located within a 5-minute travel radius.
3. Assignment: The request is sent to the nearest provider. If they do not acknowledge within 1 minute, it moves to the next nearest provider.
4. Confirmation: Once a provider accepts, the ETA is locked, and the user is notified.

Software Requirements:

Software Stack:

1. Front End: ReactJS(Vite), Tailwind CSS, shadcn/ui, Lucide React
2. Back End: Node.js, Express, Clerk Auth.
3. Database: PostgreSQL, PostGIS extension
4. Mapping: Google Maps API

Conclusion: FastPAYS aims to revolutionize the on-demand service industry by prioritizing speed and reliability. By focusing on a hyper-local algorithm, the system solves the frustration of waiting for household help, making it an essential tool for modern living.